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REPORT DOCUMENTATION PAGE

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1. REPORT NUMBER: TOP 6-2-075	2. GOVT ACCESSION NO.: A152585
3. RECIPIENT'S CATALOG NUMBER:	
4. TITLE *(and Subtitle)*: US ARMY TEST AND EVALUATION COMMAND TEST OPERATIONS PROCEDURE DISTANCE MEASURING EQUIPMENT (DME), GENERAL	5. TYPE OF REPORT & PERIOD COVERED: Final
	6. PERFORMING ORG. REPORT NUMBER:
7. AUTHOR(s):	8. CONTRACT OR GRANT NUMBER(s):
9. PERFORMING ORGANIZATION NAME AND ADDRESS: US ARMY ELECTRONIC PROVING GROUND (STEEP-TM-AC) FORT HUACHUCA, ARIZONA 85613-7110	10. PROGRAM ELEMENT, PROJECT, TASK AREA & WORK UNIT NUMBERS: AMC Reg 310-6
11. CONTROLLING OFFICE NAME AND ADDRESS: US ARMY TEST AND EVALUATION COMMAND (AMSTE-AD-M) ABERDEEN PROVING GROUND, MARYLAND 21005-5055	12. REPORT DATE: 15 March 1985
	13. NUMBER OF PAGES: 18
14. MONITORING AGENCY NAME & ADDRESS *(if different from Controlling Office)*:	15. SECURITY CLASS. *(of this report)*: Unclassified
	15a. DECLASSIFICATION/DOWNGRADING SCHEDULE:

16. DISTRIBUTION STATEMENT *(of this Report)*

Approved for public release; distribution unlimited.

17. DISTRIBUTION STATEMENT *(of the abstract entered in Block 20, if different from Report)*

18. SUPPLEMENTARY NOTES

19. KEY WORDS *(Continue on reverse side if necessary and identify by block number)*

Interrogators	VOR/TACAN/DME	Master Unit
Transponders	ICAO	Remote Unit
DME	RTCA	Rho-Theta

20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

A series of engineering tests designed to determine the technical performance of distance measuring systems and equipment is outlined in this test operation procedure (TOP). The only type of DME system treated in this TOP is the VOR/TACAN/DME.

The following tests collectively constitute an engineering evaluation of DME to determine:

(cont)

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(Item 20 continued)

a. Range and Accuracy - The average inbound lock-on and outboard unlock range limits and frequency distribution of errors within the range limits. This test is applicable to all DME systems.

b. Warmup - The time required for the test item to become operational after activation. This test is applicable to all DME systems.

c. Influence of Weather - The effects of selected meteorological conditions on the performance of the test item.

The procedures contained herein have been limited in order to provide general coverage; and designed to provide overall DME system tests that do not consider DME components in detail.

This test operations procedure applies only to the VOR/TACAN/DME distance measuring equipment described herein. Laser range finders and distance measuring equipment not employing interrogator-transponder principles are specifically excluded.

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

AMSTE-RP-702-105
Test Operations Procedure 6-2-075
AD No. A152585

15 March 1985

DISTANCE MEASURING EQUIPMENT

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1. SCOPE. A series of engineering tests designed to determine the technical performance of a distance measuring system and its equipment is outlined in this test operations procedure (TOP). The type of DME system treated in this TOP is the VOR/TACAN/DME. The following tests collectively constitute an engineering evaluation of a DME system to determine:

a. Range and Accuracy - The average inbound lock-on and outboard unlock range limits and frequency distribution of errors within the range limits. This test is applicable to all DME types.

b. Warmup - The time required for the test item to become operational after activation.

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c. Influence of Weather - The effects of selected meteorological conditions on the performance of the test item.

1.1 Limitations

The procedures contained herein have been limited in order to provide general coverage; and designed to provide overall DME system tests that do not consider DME components in detail. Laser range finders and distance measuring equipment not employing interrogator-transponder principles are specifically excluded. This test operations procedure applies only to the VOR/TACAN/DME system.

1.2 Background

a. The VOR/TACAN/DME system is defined by MIL-STD-291A, (ref 4B) ICAO Annex 10 (ref 4C), and the RTCA Minimum Performance Standard (ref 4D). It is a pulsed interrogator-transponder system operating in the 960-1215 MHz band, and is common to three rho-theta (ρ-θ) navigation systems: the military TACAN system, the VORTAC system, and the VOR/DME system (all depicted in figure 1). Specifications for the TACAN system were established in 1951, for the VORTAC in 1956, and for the VOR/DME system in 1959.

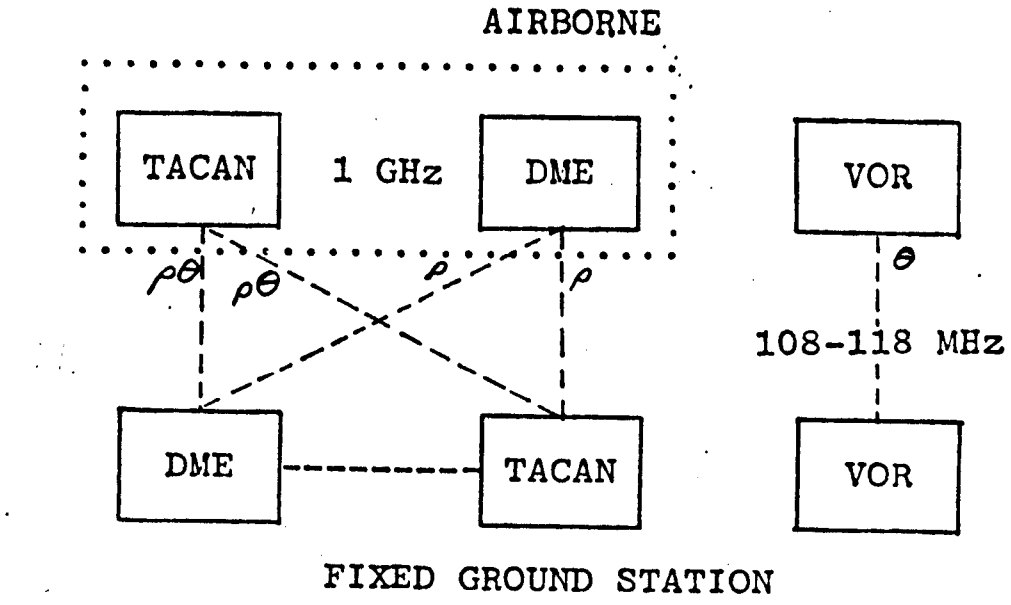


Figure 1. The VOR/TACAN/DME system.

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b. The remote unit receives signals from the master unit and amplifies and retransmits them. The phases of the returned signals at the master unit are compared with the phases of the outgoing signals, thereby obtaining a measure of signal distance covered. It is necessary to perform engineering tests of general distance measuring equipment to (1) determine performance relative to test criteria, (2) define safety aspects, and (3) determine suitability for operational testing. A listing of textual terms and abbreviations peculiar to DME sytems is presented in the glossary at the end of this TOP.

2. FACILITIES AND INSTRUMENTATION. Place the test item in operating condition as outlined in the equipment technical manual.

2.1 Facilities.

a. Army aircraft with complete operating systems.

b. Test range with ground station equipment systems.

c. Bench test facilities complete with power supplies, equipment, and instrumentation.

d. Photographic and video recording equipment.

e. Meteorological support facilities and equipment.

2.2 Instrumentation.

a. Absolute altimeter.

b. Barometric altimeter.

c. Radio frequency aircraft tracking instrumentation.

d. Very high frequency omnidirectional radio range (VOR) equipment.

e. Onboard instrumentation recording capabilities.

f. Radar or cinethodolite tracking instrumentation.

g. VOR/DME or VORTAC ground station transponders.

h. Meters, oscilloscopes, counters, analyzers, etc., as required.

3. PREPARATION FOR TEST

a. Select test equipment having an accuracy of at least ten times that of the function to be measured.

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b. Record the following information:

(1) Nomenclature, serial number(s), and the manufacturer's name of the test item(s).

(2) Nomenclature, serial number, accuracy tolerances, calibration requirements, and last date calibrated of the test equipment selected for the tests.

c. Assure that all test personnel are familiar with the required technical and operational characteristics of the item under test, such as stipulated in the Materiel Need (MN), the Required Operational Capability (ROC), the Qualitative Material Requirements (QMR), Small Development Requirements (SDR), Technical Characteristics (TC), Letter Requirement (LR), and Letter of Agreement (LOA).

d. Review all instructional material issued with the test item by the manufacturer, contractor, or government, as well as reports of previous similar tests conducted on the same type of items, and familiarize all test personnel with the contents of such documents. These documents shall be kept readily available for reference.

e. Thoroughly inspect the test items for obvious physical and electrical defects such as cracked or broken parts, loose connections, bare or broken wires, loose assemblies, bent relay and switch springs, corroded jacks and plugs, and bare or cracked insulation. All defects shall be noted on an applicable data form, and corrected before proceeding with the tests.

f. Prepare record forms for systematic entry of data, chronology of test, and analysis in final evaluation.

g. Prepare adequate safety precautions to provide safety for personnel and equipment, and ensure that all safety precautions are observed throughout the test.

h. Ensure that adequate maps and navigation charts [such as DOD Flight Information Publication (FLIP) or other navigation charts] are available for testing of airborne DME units.

i. Ensure that adequate test range radar support is available for all airborne testing.

j. Ensure that line-of-sight ranges with accurately measured distances between established points have been established and are available for tests as required.

k. Mount the following equipment in the test aircraft:

(1) Test item (DME interrogators)

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(2) Transponder beacon (such as the AN/DPM-131) to enhance the range radar's return signal

(3) Cameras to record DME readout, altitude, and time

(4) Very high frequency omnidirectional radio ranging equipment (VOR) for determining bearings

l. Require airworthiness releases to ensure that the test item does not prevent safe operation of the aircraft.

m. Investigate the possible adverse influence of the test item on the aircraft or its subsystems. Develop procedures for conducting interface testing between the test item and the aircraft.

4. TEST CONTROLS

a. Organize test team and establish responsibilities for test conduct, reporting, and data control.

b. Select test equipment having an accuracy at least one order of magnitude greater than that of the function to be measured.

c. Familiarize all test personnel with the test item instructional material.

d. Prepare adequate safety precautions to provide safety for personnel and equipment and ensure all safety SOPs are observed throughout the test.

e. Inspect the test item thoroughly for obvious physical and electrical defects. Note and correct all defects before proceeding with testing.

f. Prepare, disseminate, and continually review record forms for systematic entry of data, chronology of tests, and analysis in final evaluation of the test item.

g. Prepare and monitor a test item sample plan sufficient to ensure that enough samples of all measurements are taken to provide statistical confidence of final data in accordance with TOP 3-1-002, Confidence Intervals and Sample Size, 25 January 1967, AD No. 718229.

h. Perform such preliminary tests as necessary to assure the test item(s) is in satisfactory condition.

i. Obtain continuous environmental and meteorological data for the entire test area during the test program from the meteorological support activity or from appropriate instrumentation.

j. Monitor and record transmissions from the master and slaves during the system test with monitor equipment for comparison purposes.

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5. PERFORMANCE TESTS

5.1 Range and Accuracy

5.1.1 VOR/TACAN/DME

NOTES: 1. The VOR/TACAN/DME range and accuracy test shall be conducted in a test where range radar coverage is available and where a VOR/DME or VORTAC transponder can be used as the hub for the flight test courses.

2. The flight test shall consist of approximately six radial flight paths centered upon the ground transponder as illustrated in figure 2. The radials shall be spaced at equal angles insofar as possible to minimize "siting" effects.

a. Select the test course, determine (with the aid of the equation or nomographs given in appendix A) the approximate maximum range at which testing shall be initiated.

b. Direct the aircraft containing the test item to fly over the test course at a velocity of approximately 50 percent of the aircraft's rated ground speed and at an inital altitude of 1500 feet above the ground station.

c. Direct the test pilot to fly each leg of the test course in each direction. The pilot shall continuously interrogate the ground to determine the points at which inbound lock and outbound unlock occur. On-board recording equipment shall monitor and record DME readout, altitude, pitch, roll, yaw, and time synchronization marks. The test pilot shall introduce varying degrees of aircraft pitch, roll, and yaw during the test in order to evaluate test item performance as a function of aircraft attitude.

d. Record numerous readouts between the lock and unlock points to provide statistical data on the frequency distribution of DME error. The range radar station shall provide time correlated space positions of the aircraft.

e. Repeat the above procedure for aircraft altitudes of 3000, 6000, 9000, 12000, and 15000 feet and additional intermediate and extreme values as dictated by the test item's specifications.

f. Repeat the entire test sequence as necessary to provide a sufficient number of measurements for drawing statistically valid conclusions. The background document TOP 3-1-002, Confidence Intervals and Sample Size, should be consulted for background information and details.

5.2 Warmup

NOTE: The warmup test shall be conducted in an open field configuration with the interrogator/master unit bench mounted as convenient, and the transponder/remote unit located a sufficient distance away so as to achieve a convenient mid-scale distance reading.

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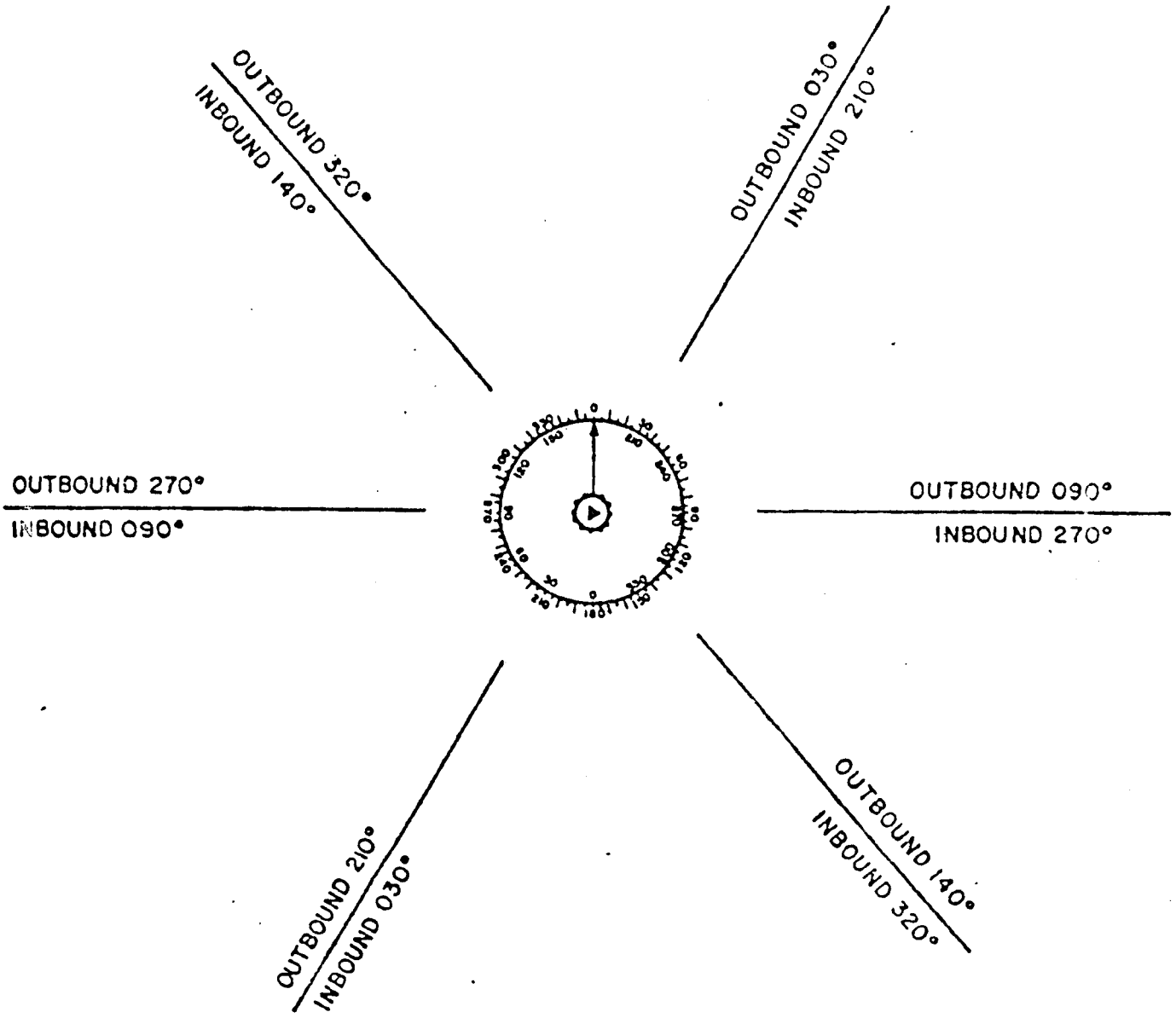


Figure 2. Typical airborne DME test flight paths.

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a. Ensure that the interrogator/master unit has been in the "off" position a sufficient length of time for it to reach thermal equilibrium with the environment and, conversely, ensure that the transponder/remote unit has been in the "on" position a sufficient length of time to achieve stable operation.

b. Turn on the interrogator/master unit and note the elapsed time between equipment turn-on and achievement of stable distance measurement readout.

c. Repeat the above procedure a minimum of five times in order to be able to compute the mean warmup time.

d. Repeat steps a through c above at the lowest and highest ambient temperatures available without the use of an environmental chamber.

e. Repeat the entire test sequence listed above in the following variations:

(1) Ensure that the transponder/remote unit is in thermal equilibrium with the environment and the interrogator/master unit has achieved stable operation and thereby reverse the initial test conditions.

(2) Ensure that both the transponder/remote unit and the interrogator/master unit are in thermal equilibrium with the environment, and simultaneously turn on both units.

5.3 Warning and Restricting Devices. The engineering evaluation of the test item's warning and restricting devices shall be performed as an open field test with the airbone interrogator unit either bench mounted or mounted in a parked aircraft. The transponder unit shall be placed a sufficient distance from the integrator to achieve a convenient on-scale distance reading. The transponder unit operator shall (1) attempt voice transmission during interrogation and reply and (2) complete the following questionnaire following the attempt:

a. Does the transponder unit positively indicate when it is being interrogated?

Yes ( ) No ( ) Comment _____

b. Does the transponder unit positively indicate when it is replying to an interrogation?

Yes ( ) No ( ) Comment _____

c. Is the design of the transponder unit such that it prevents voice transmission during interrogation or reply?

Yes ( ) No ( ) Comment _____

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d. If voice transmission is possible during interrogation or reply, is there a detrimental effect on the distance readout?

Yes ( ) No ( ) Comment _____

5.4 Influence of Weather. The influence of varying degrees of meteorological phenomena on the range and accuracy of the test item shall be determined by repeating the procedures outlined in paragraph 5.1 under the following weather conditions:

a. Heavy rainfall

b. Dust

c. Lightning

d. Operation over hot, sun-heated surfaces

NOTE: Quantitative assessment of the net effect of varying intensities of the above conditions will be difficult to achieve due to the inherent problem of locating the desired weather phenomenon within the test area. If this problem can be overcome, the very occurrence to which the test item is subjected, such as heavy rainfall, may obscure the aircraft from the tracking system. Because of these problems, collection of a sufficient number of data points to allow a statistical analysis may not be practical and all airborne test items may have to be tested in a ground survey manner as indicated in paragraph 5.1.3.

6. DATA REQUIRED

6.1 Preparation for Test. Data to be recorded prior to testing will include but not be limited to:

a. Nomenclature, serial number(s), and manufacturer's name(s) of the test item(s), and all components and accessories.

b. Nomenclature, serial number, accuracy tolerances, calibration requirements, and last date calibrated of the test equipment selected for the tests.

c. Defects noted as a result of pre-test inspection.

d. Prevailing meteorological conditions prior to start of test as well as during test conduct, to include:

(1) Temperature

(2) Humidity, relative or absolute

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(3) Temperature gradiant

(4) Atmospheric pressure

(5) Precipitation

(6) Solar radiation

(7) Wind speed and direction

(8) Pibal (Pilot balloon, winds aloft) data

(9) Frequency of readings

(10) Source of data

6.2 Test Conduct

6.2.1 Range and Accuracy Test Data

6.2.1.1 VOR/TACAN/DME. Range and accuracy test data for VOR/TACAN/DME test items shall consist of:

a. DME readout vs time

b. Time correlated space position data

c. Meteorological conditions

d. Aircraft altitude; degree of pitch, roll, and yaw

e. Test course identification

6.2.2 Warmup. Warmup test data to be recorded shall be:

a. Ambient temperature

b. Interrogator/master unit warmup time

c. Transponder/remote unit warmup time

d. Simultaneous warmup time

6.2.3 Warning and Restricting Devices. Warning and restricting devices test data shall consist of answers to the questionnaire given in paragraph 5.3, together with pertinent comments and/or clarifying remarks.

6.2.4 Influence of Weather. Influence of weather test data to be recorded shall consist of values of those parameters indicated in paragraph 6.2.1.

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7. DATA REDUCTION AND PRESENTATION. Processing of raw test data shall, in general, consist of:

a. Reducing time correlated space position data to distance-to-go (slant range from interrogator/master unit to transponder/remote unit).

b. Applying statistical theory to:

(1) Determine mean values and frequency distribution of errors.

(2) Account for error introduced by the measurement system.

c. Organizing data into tabular and graphical form.

All test data shall be properly marked for identification and correlation and grouped according to subtest title. Specific instructions for the reduction and presentation of individual subtest data are outlined in the succeeding paragraphs.

7.1 Range and Accuracy

7.1.1 VOR/TACAN/DME. Graphic presentations of the VOR/TACAN/DME range and accuracy test shall be made as follows:

a. Terrain map of test area showing:

(1) Latitude and longitude of ground transponder station

(2) Test flight courses

(3) Ground transponder indentification and modulation information

b. Mean value inbound lock-on and outbound unlock curves as shown in figure 3.

c. Standard deviation from mean value inbound lock-on and outbound unlock curves either plotted on the same graph as the mean value curves or on a similar graph as indicated in figure 3.

d. Extreme value inbound lock-on and outbound unlock curves similar to that shown in figure 3.

e. Frequency distribution of errors curve as shown in figure 4.

7.2 Warmup. Test data relative to warmup shall be reduced to mean values of time for each of the three conditions indicated in paragraph 6.2.2 and presented as a numerical value that can be correlated to ambient temperature.

7.3 Warning and Restricting Devices. The test data collected from the warning and restricting devices shall be an organized presentation of the data as recorded under paragraph 6.2.3.

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7.4 Influence of Weather. Test data from the influence of weather test shall be presented according to paragraph 7.1.

A written report shall accompany all test data and shall consist of conclusions and recommendations drawn from test results. The test engineer's opinion, concerning the success or failure of any of the function evaluated, shall also be included. In addition, equipment specifications that will serve as the model for a comparison of the actual test results should be included.

Equipment evaluation usually will be limited to comparing the actual test results to the equipment specifications and the requirements as imposed by the intended usage. The results may also be compared to data gathered from previous tests of similar material.

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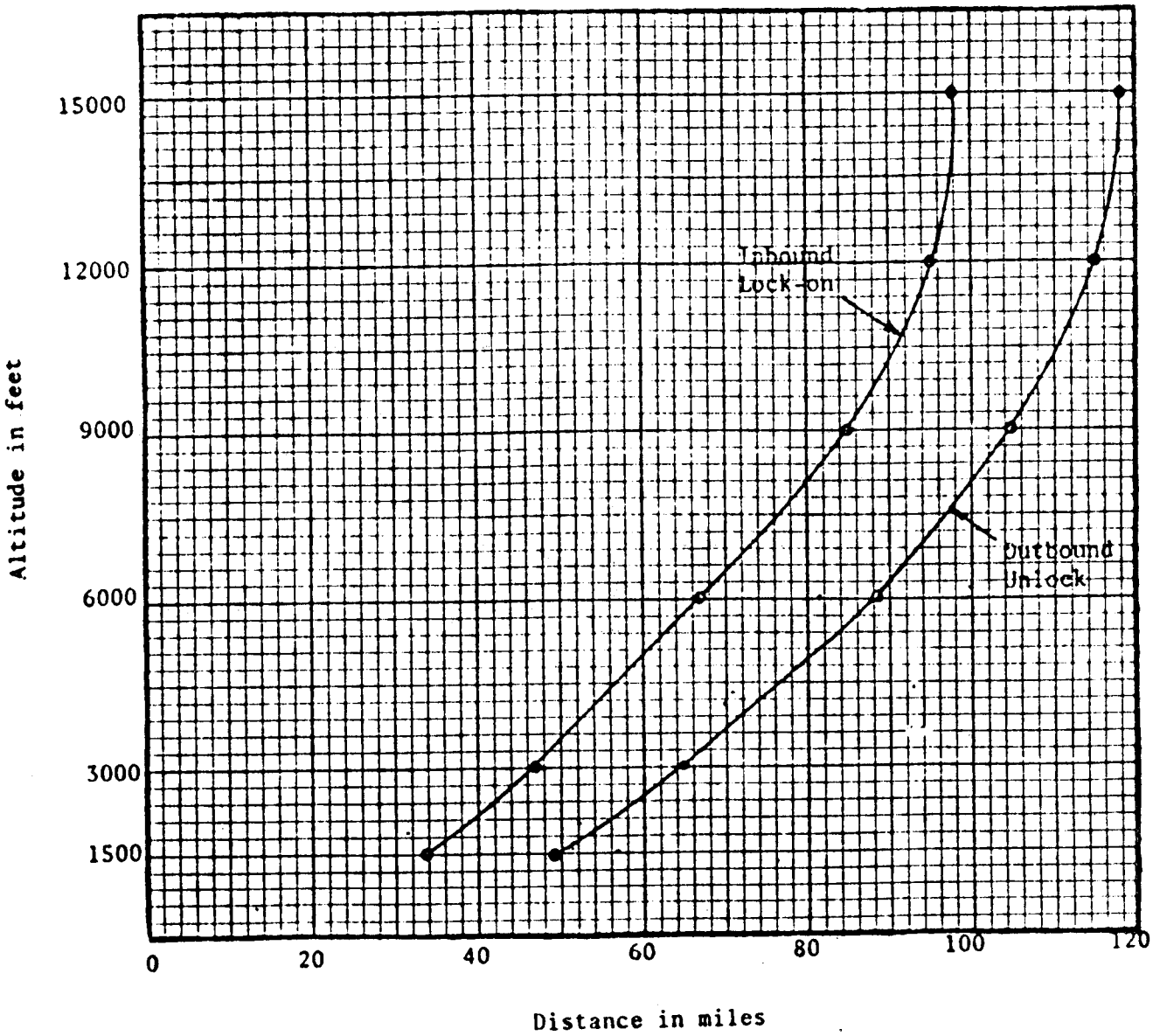


Figure 3. Mean value inbound lock-on and outbound unlock curves.

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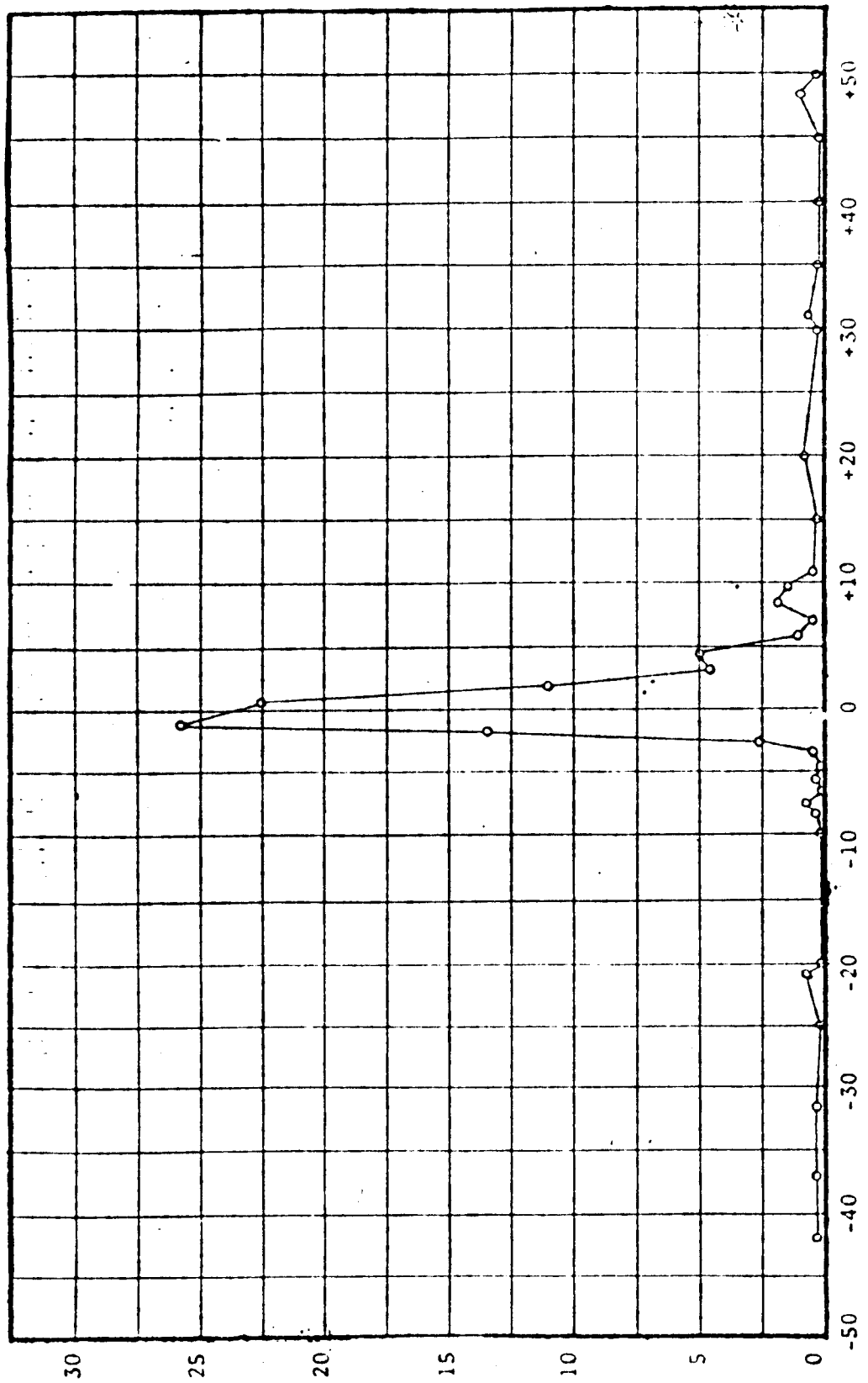


Figure 4. Frequency distribution of errors of all DME interrogations.

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Numerical range and accuracy data shall be reduced to and presented as tabular listings similar to the sample in figure 5.

Test Course: _____ (e.g., 320° outbound radial)

Altitude: _____

Meteorological Conditions: _____

DME Reading (NMI)	Slant Range (NMI)	Difference ±	Specified Tolerance (NMI)	Aircraft Altitude Pitch, Roll, & Yaw

Figure 5. Numerical range and accuracy data.

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Appendix A. APPROXIMATION OF MAXIMUM DME RANGE

The point at which a radio wave traveling directly from a transmitter to a receiver first touches the surface of the earth may be calculated and used as an approximation of maximum DME range. This maximum radius for testing is determined by the expression:

$$d_L = \sqrt{2h_1} + \sqrt{2h_2}$$

(the bending of the ray due to atmospheric refraction is accounted for)

where: h_1 = transmitting antenna height in feet

h_2 = receiving antenna height in feet (aircraft altitude)

and: d_L = radio line-of-sight distance between antennas in miles (radio horizon)

A nomograph for this calculation is shown in figure A-1.

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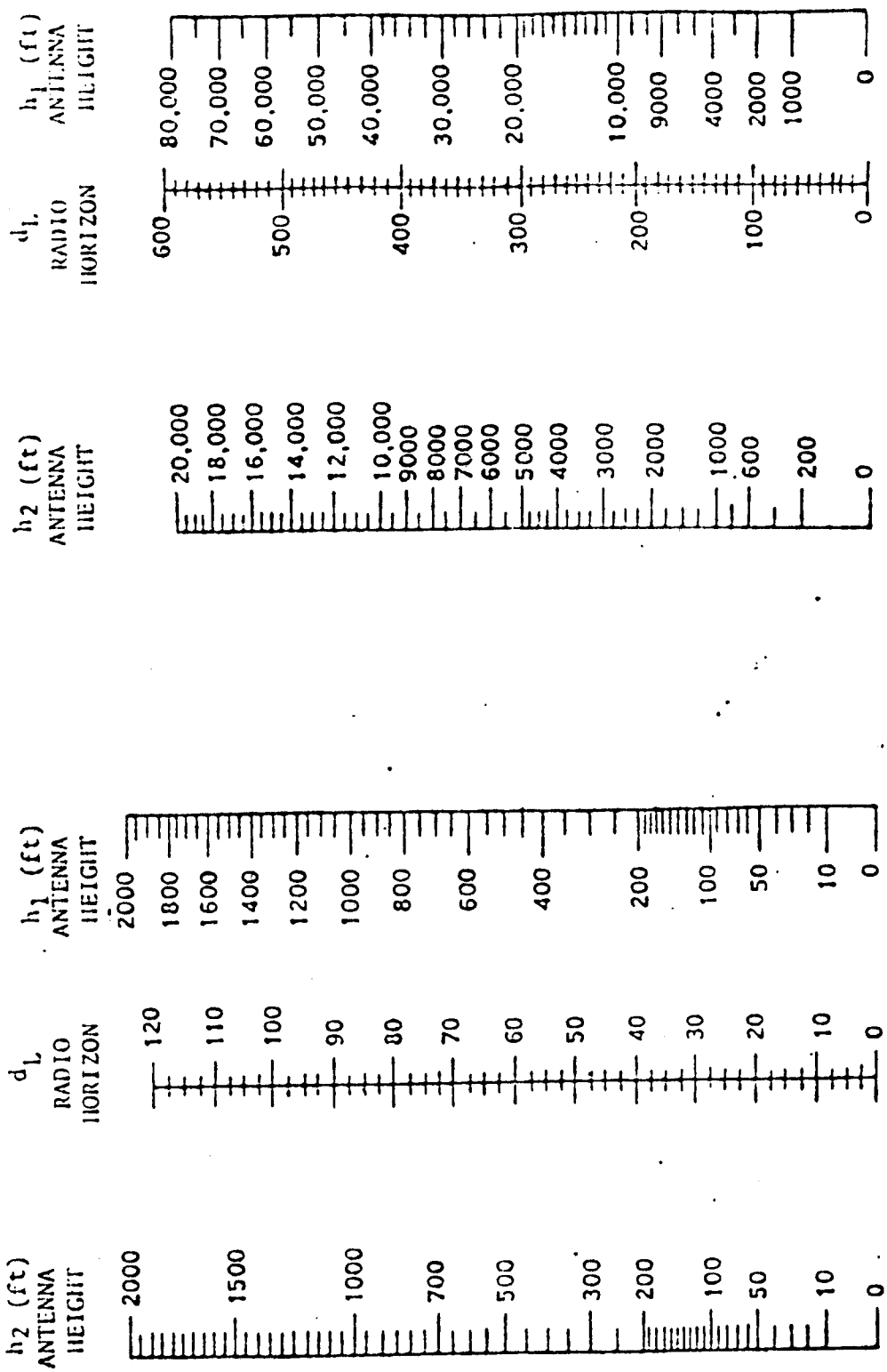


Figure A-1. Nomographs giving radio horizon distance.

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GLOSSARY

DME	Distance measuring equipment
VOR	Very high frequency omnidirectional radio range
TACAN	Tactical air navigation
VHF	Very high frequency
FM	Frequency modulation
ICAO	International Civil Aviation Organization
RTCA	Radio Technical Commission for Aeronautics
Rho-Theta	(ρ-θ) range - bearing
Interrogator	The airborne portion of the VOR/TACAN and VHF-FM distance measuring equipment
Transponder	The ground station portion of the VOR/TACAN and VHF-FM distance measuring equipment
Master Unit	That portion of the tellurometer distance measuring system which supplies the initial measuring frequency and the final distance readout
Remote Unit	That portion of the tellurometer distance measuring system which relays the initial measuring frequency back to the master unit
FLIP	Flight Information Publication
Pibal data	Pilot Balloon, winds aloft data
NMI	nautical miles

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